ISyE 3770 HW2

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Problem 2.5.4

Part A:

We find the probability that the failure involves a gas leak by calculating the sum of all failures involving a gas leak over the total number of failures. Therefore, we have:

$$\frac{55+32}{55+32+17+3} = \frac{87}{107}$$

```r
87 / 107
```

```
## [1] 0.8130841
```

Part B:

We search for $P(A|B)$, where A is evidence of electrical failure and B is a gas leak. Therefore, we know by the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{55}{55+32}$$

```r
55/(55+32)
```

```
## [1] 0.6321839
```

Part C:

Similarly to Part Two, we now search for $P(B|A)$, using A and B as defined above.

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{55}{55+17}$$

```r
55/(55+17)
```

```
## [1] 0.7638889
```

Problem 2.5.6

Part A:

We wish to find $P(B|A)$, where A is the first OJ selected being defective and B is the second one selected is defective. Therefore, we know $P(A) = \frac{5}{500}$. Similarly, find $P(B|A) = \frac{4}{499}$ since there exists 499 possible OJ containers left with 4 being defective.

```r
4/499
```

```
## [1] 0.008016032
```

Part B:

We wish to find the probability of A and B. Therefore, we have $P(A \cap B) = P(A) * P(B|A) = \frac{5}{500} * \frac{4}{499}$.

```r
(5/500)*(4/499)
```

```
## [1] 8.016032e-05
```

Part C:

We wish to find the probability that the first two are acceptable. Therefore, we wish to find $P(A' \cap B') = P(A') * P(B'|A') = \frac{495}{500} * \frac{494}{499}$

```r
(495/500)*(494/499)
```

```
## [1] 0.9800802
```

Part Extra:

The probability of the second one being defective is expressed as $P(B) = P(B \cap A) + P(B \cap A') = P(B|A) * P(A) + P(B|A') * P(A')$. Using previously found values, we have $P(B) = \frac{4}{499} * \frac{5}{500} + \frac{495}{500} * \frac{5}{499}$.

```r
(4/499)*(5/500)+(495/500)*(5/499)
```

```
## [1] 0.01
```

Problem 2.6.3

We will define A as the state of being dry and B as the electrical connector failing. Therefore, we are searching for $P(B)$ which we find using the following:

$$P(B) = P(B|A) * P(A) + P(B|A') * P(A')$$

We know $P(A) = 0.90$ and $P(A') = 0.10$ from the problem. Furthermore, we have $P(B|A) = 0.01$ and $P(B|A') = 0.05$. Therefore, we find: $P(B) = 0.01(0.90) + 0.05(0.10)$

```r
(0.01*0.90)+(0.05*0.10)
```

```
## [1] 0.014
```

Problem from Notes

Requirements: 8 characters, 26 lowercase, 26 uppercase, 10 integers

Part A:

To find $P(A|B')$, we must first find $P(B') = 1 - P(B) = 1 - \frac{10^8}{62^8}$. Therefore, we find: $P(A|B') = \frac{P(A \cap B)}{P(B')} = \frac{P(A)}{P(B')} = \frac{\frac{52^8}{62^8}}{1 - \frac{10^8}{62^8}}$

```r
(52^8/62^8)/(1-(10^8/62^8))
```

```
## [1] 0.2448462
```

Part B:

We find $P(A' \cap B) = P(B)$ since B and not A is B. Therefore, we find: $P(B) = \frac{10^8}{62^8}$

```r
10^8/62^8
```

```
## [1] 4.580011e-07
```

Part C:

We wish to find $P(\text{password contains exactly 2 integers given that it contains at least 1 integer})$. We will let A represent exactly two integers and B represent at least one integer. Therefore, we have the form $P(A|B')$. We wish to solve $\frac{P(A \cap B')}{P(B')}$. We know $P(A \cap B') = P(A)$ since a password containing two integers already contains at least one integer. Additionally, we find $P(B') = 1 - P(B) = 1 - \frac{52^6}{62^6}$. Therefore, we have: $\frac{P(A)}{P(B')} = \frac{\frac{\binom{8}{2}*10^2*52^6}{62^8}}{1-\frac{52^8}{62^8}}$.

```r
((choose(8,2)*10^2*52^6)/(62^8))/(1-(52^8/62^8))
```

```
## [1] 0.3357447
```

Problem 2.7.4

To determine if events A and B are independent, we must prove the following:

$P(A|B) = P(A)$

$P(B|A) = P(B)$

$P(A \cap B) = P(A) * P(B)$

We begin by attempting to prove the first statement. We have the following: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{70}{79}$. We find $P(A) = \frac{86}{100}$.

```r
70/79
```

```
## [1] 0.8860759
```

```r
86/100
```

```
## [1] 0.86
```

Therefore, sine $P(A|B) \neq P(A)$, we find the two events A and B are not independent.

Problem 2.8.9

Part A:

We find $P(free) = 0.20(0.60) + 0.80(0.04)$

```r
(0.20*0.60)+(0.80*0.04)
```

```
## [1] 0.152
```

Part B:

We wish to find $P(spam|free)$. Therefore, using Bayes Theorem, we have: $P(spam|free) = \frac{P(free|spam)*P(spam)}{P(free)} = \frac{0.60*0.2}{0.20(0.60)+0.80(0.04)}$

```r
(0.60*0.2)/(0.20*0.60+0.80*0.04)
```

```
## [1] 0.7894737
```

Part C:

We wish to find $(P(spam'|free'))$. We will similarly utilize Bayes Theorem and produce: $(P(spam'|free')) = \frac{P(free'|spam')*P(spam')}{P(free')} = \frac{(1-0.04)*0.80}{1-0.152}$

```r
(1-0.04)*0.08/(1-0.152)
```

```
## [1] 0.09056604
```

Question Two

Given A and B are independent, we will prove A' and B' are also independent.

$$P(A \cap B) = P(A) * P(B) = P(A) + P(B) - P(A \cup B)$$

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

By definition of the inverse, we have:

$$P(A') = 1 - P(A) and P(B') = 1 - P(B)$$

$$P(A')P(B') = 1 - P(A) - P(B) + P(A)P(B)$$

Therefore, we have:

$$P(A' \cap B') = P(A')P(B')$$

And so, by definition of independence, we know A' and B' are independent.